Foundational Models

❖ What it is:

- **Foundational models are the AI models which are trained on large amount of data to do a wide range of downstream tasks (the task that we want to solve), such as sentiment analysis of text, question answering, text summarization, image generation etc.**
- **They can work with text, images, audio, video, code and other type of data.**
- **Imagin building an AI assistant, we can build a separate model for answering questions, another for summarizing documents, and another for generating text. That could be a lot of unnecessary work. Instead of that, we can start with a model trained on broad range of data and adapt it for different tasks.**

❖ Working:

- **Pre-training:**
  - **The model is trained using self-supervised learning technique (SSL).**
  - **In SSL, model is trained using data that does not have any labels or answers provided. Instead of needing people to label data, the model finds the patterns and creates its own labels from data automatically i.e. model derives targets from unlabelled data.**
  - **SSL doesn't mean there is no human intervention, it means that humans does not need to manually label every training example with answer (target).**
  - **The model trains itself by playing endless guessing games. It hides the piece of data from itself and tries to predict it.**
  - **For example, consider the sentence "The sky is blue." During training, the training process may hide the word "blue" and ask the model to predict the missing word from the surrounding context. Suppose in first iteration model predicts the word "green" (which is a wrong), training process calculates the loss and backpropagates then adjust the parameters and predicts again. This process continues till model improves its predictions.**
  - **This allows the model to learn useful information by teaching itself from the data.**
  - **For text, it hides the word and guess it based on the context.**
    For image, it hides the patch of an image and try to reconstruct what is missing.

- **Adaptation:**
  - **A pre-trained model can be adapted to specific requirement.**

- This is done by Fine-Tuning.
- Fine tuning is the process of continuously training a foundational model on targeted, domain-specific dataset to permanently alter its behaviour and knowledge.
- Example includes fine tuning a foundational model for medical field, finance, customer support etc.
- The benefit of fine tuning is that we don't have to train a model from scratch, we take an already trained model which can understand human language.

- **Deployment:**
  - The adapted i.e. fine-tuned model is integrated into an application such as a chatbot, image-generation system, coding assistant, customer support, or other AI-powered application.

❖ **Types of FMs:**

- **Large Language Models (LLMs):**
  - These models are primarily trained on text and can be adapted for tasks such as text generation, translation, summarization, chatbot interactions, and question answering etc.
  - Example: BERT, GPT, Claude, and Llama.
- **Vision Foundation Models:**
  - These models are trained on image datasets to perform tasks like image generation, object detection, and image classification.
  - These models can be fine-tuned for specific applications, such as medical image analysis or object recognition in autonomous vehicles.
  - Another capability is the generation of an images from text.
  - Example: DALL-E(OpenAI), Stable diffusion (Stability AI), Imagen (Google).
- **Code Generation Models:**
  - These models can assist with completing, debugging, and generating code in different programming languages.
  - Example: OpenAI Codex, Code Llama (Meta), StarCoder (BigCode), CodeGemma (Google).
- **Multimodal Foundational Models:**
  - These models are trained to process diverse data types, such as text, images, audio, or video.
  - Example: Gemini (Google), GPT-6 Astra (Open AI), Claude (Anthropic), Llama 4 (Meta).

❖ How To Pick Right AI Foundation Model:

- **Step 1: Articulate Use case**
  - First, clearly define the problem organization want to solve using AI.
  - This will help to narrow down the suitable models.
  - For example, an organization may need an AI model to summarize documents, generate code, answer customer queries etc.
- **Step 2: List Model Options**
  - Identify the available foundation models that could perform required tasks.
  - The selection should focus on model's relevancy to the use case.
- **Step 3: Identify Model Characteristics**
  - Compare selected models based on their characteristics, including:
    - Model Size: Number of parameters in the model
    - Performance: Ability to complete the required task accurately
    - Cost: API usage and Token charges, computation resources maintenance expenses
    - Risks: Potential bias, hallucinations, security concerns, data privacy leaks.
- **Step 4: Refine Selection**
  - Refine the shortlist by checking how well each model's capabilities align with specific requirements of the use case.
  - For example, if an organization wants to build a code generation application, it should prioritize models with strong coding capabilities rather than selecting general purpose models.
- **Step 5: Testing**
  - Test the shortlisted Foundational Model using representative inputs to evaluate their performance against:
    - Accuracy: How correctly the model performs the required tasks
    - Reliability: How consistently the model produces the appropriate results
    - Speed/Latency: How quickly or slowly the model responds
  - Compare the test results to identify each model's strengths and weaknesses.
- **Step 6: Choose the best value**
  - Select the Foundational Model that offers best balance of accuracy, reliability, speed, cost and security.
  - The largest model is not always the best choice as smaller models may meet the requirements at lower cost.